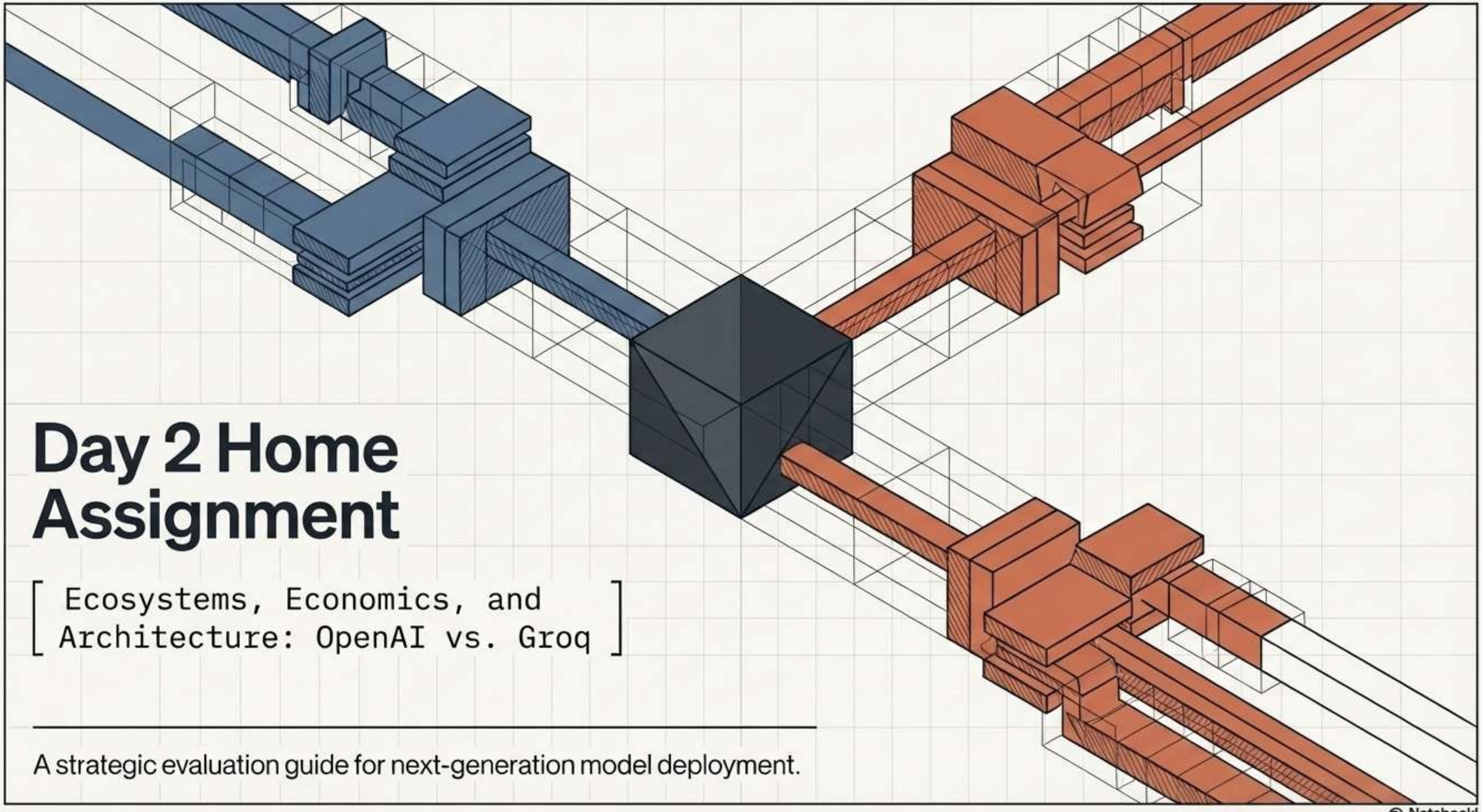


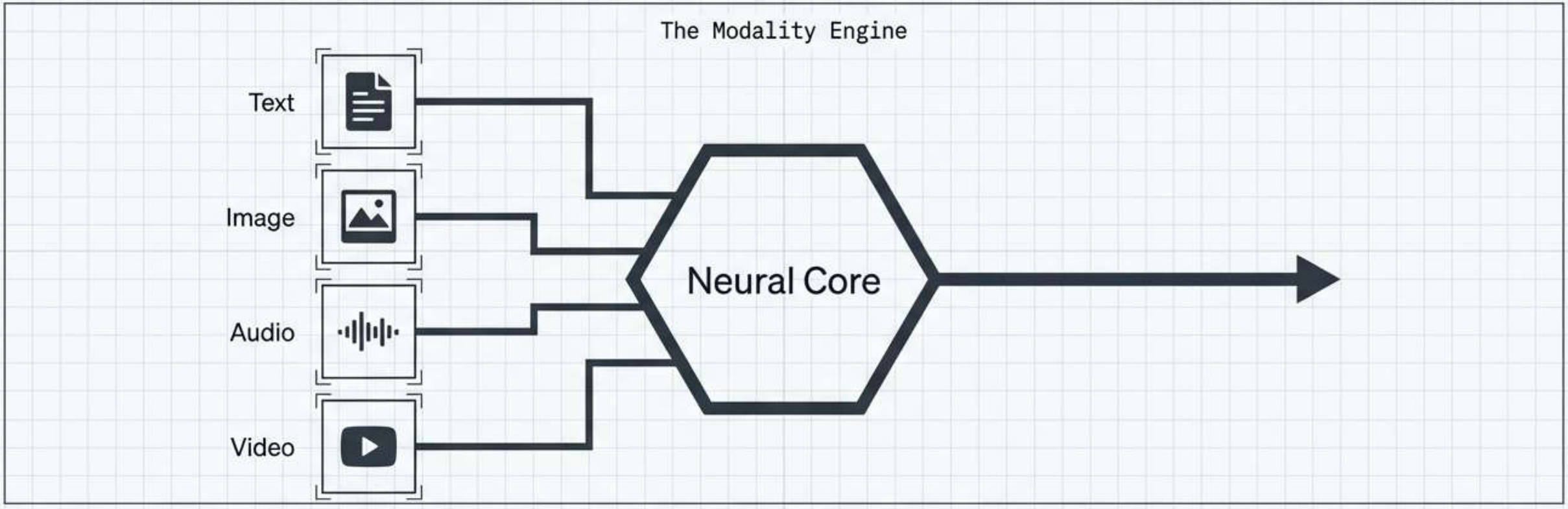
Day 2 Home Assignment

[Ecosystems, Economics, and Architecture: OpenAI vs. Groq]

A strategic evaluation guide for next-generation model deployment.

NotebookLM

Multimodal architectures transcend text to integrate diverse sensory inputs into a unified reasoning engine.



Multimodal AI refers to machine learning models capable of processing and integrating information from multiple modalities or types of data.

> STATUS_UPDATE: The landscape has definitively shifted from unimodal language generators to comprehensive, multi-sensory models.

NotebookLM

Open AI models

GPT-4.1

The smartest non-reasoning model remaining in the active ecosystem.

GPT-5 Mini

Near-frontier intelligence built for cost-sensitive, low-latency, and high-volume workloads.

GPT-5 Nano

The fastest, most cost-efficient version of the GPT-5 generation.

GPT-5 Base

The previous intelligent reasoning model for coding and agentic tasks, featuring configurable reasoning effort.

NotebookLM

Groq models

LLaMA (3.3 / 4)

The Heavyweight. Provides massive open-source scale and reasoning capabilities.

Mistral

The Agile Enterprise. Features Mistral Large for strong enterprise routing and Mistral Small for ultra-low-cost execution.

Whisper

Modality focus: High-fidelity audio transcription and processing at ~$0.006/min.

CLIP

Modality focus: Vision and image integration for visual reasoning tasks.

NotebookLM

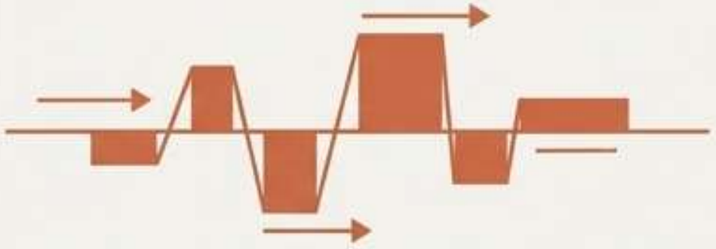
Classification of Models

Model	Context Window	Input Cost (per 1M)	Output Cost (per 1M)	Key Differentiator
Mistral Small	~32K–128K	~$0.05 – $0.20	~$0.08 – $0.60	Ultra low cost
LLaMA (3.3 / 4)	~128K (up to 1M)	~$0.18 – $1	~$0.40 – $2	Cheapest high-scale option
GPT-5.4 Mini	~128K–200K	~$0.5 – $2	~$1 – $5	Cheapest OpenAI tier
Mistral Large	~128K	~$0.5 – $2	~$1.5 – $6	Strong enterprise model
GPT-5.4	~200K–1M	~$5 – $10	~$15 – $30	Balanced intelligence
GPT-5.4 Pro	~1M+	~$10 – $20	~$30 – $60	Deep reasoning
Whisper	N/A	N/A	~$0.006/min	Audio processing

NotebookLM

Model deployment requires precise workflow-matching to prevent catastrophic margin erosion.

The Efficiency Maximizer

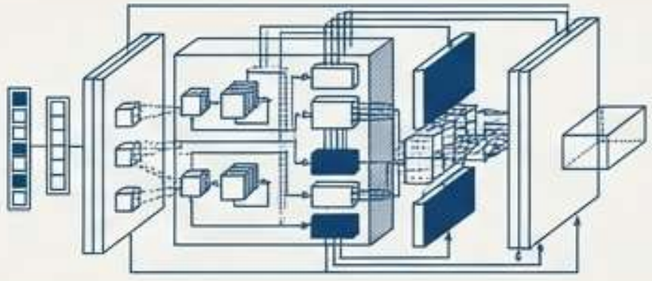


Mistral Small

- ~$0.05 Input / ~$0.08 Output
- Ideal for massive-scale, simple log-parsing or basic routing.

300x TO 750x COST MULTIPLIER

The Deep Reasoning Engine



GPT-5.4 Pro

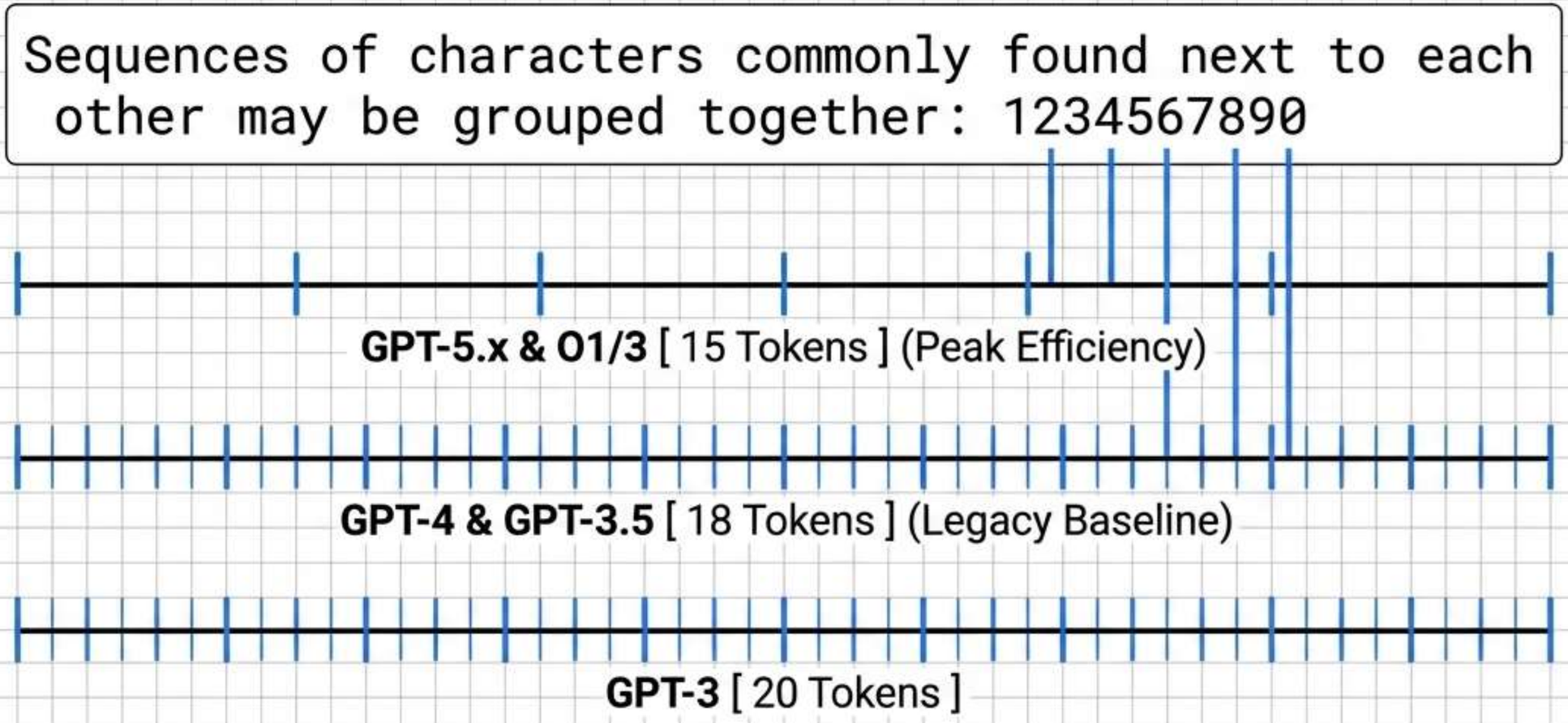
- ~$10-20 Input / ~$30-60 Output
- 1M+ Context Window
- Reserved exclusively for complex, multi-step agentic workflows.

Treating model selection as a 'one-size-fits-all' procurement guarantees either architectural failure or extreme financial waste.

NotebookLM

Tokenizer

Tokenizers group sequences of characters commonly found next to each other.



NotebookLM